STM32F103 MICROCONTROLLER PRINTED CIRCUIT BOARD DESIGN

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AKOKA, NIGERIA

APRIL, 2024

1.0 Background Study

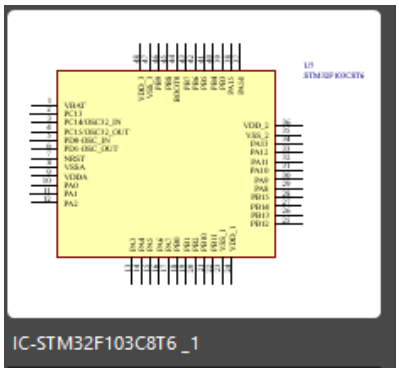
The STM32F103 is a versatile and widely-used 32-bit microcontroller for embedded development, offering a balance of performance, features, and cost-effectiveness. It offers high performance and low power consumption, making it suitable for a wide range of embedded applications. The STM32F103 series features a rich set of peripherals including GPIO (General Purpose Input/Output) pins, UART (Universal Asynchronous Receiver/Transmitter), SPI (Serial Peripheral Interface), I2C (Inter-Integrated Circuit), ADC (Analog-to-Digital Converter), timers, PWM (Pulse Width Modulation), and more. The STM32F103 microcontroller can be programmed in STM32CubeIDE and C/C++ using libraries provided by STMicroelectronics or third-party libraries.
Due to its performance, peripherals, and low-power features, the STM32F103 microcontroller is commonly used in a wide range of embedded applications including industrial control systems, consumer electronics, smart appliances, medical devices, automotive systems, and more.

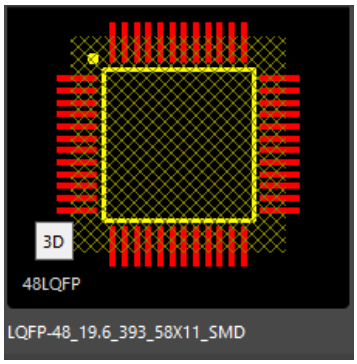
2.0 Aim and Objectives

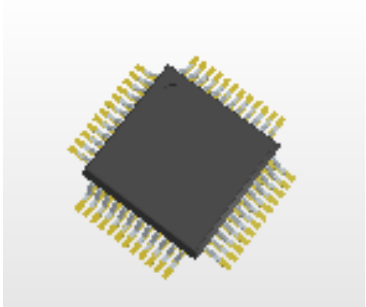
To design a Printed circuit Board of STM32F103 Microcontroller using Altium designer

3.0 Apparatus/Component Used

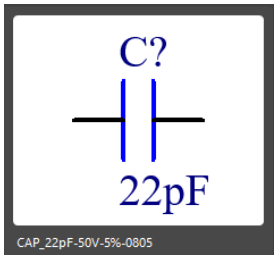
1. STM32F103C8T6 Microcontroller

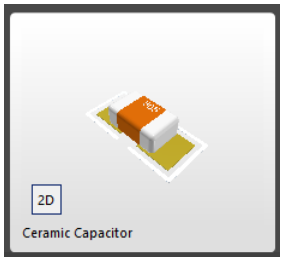


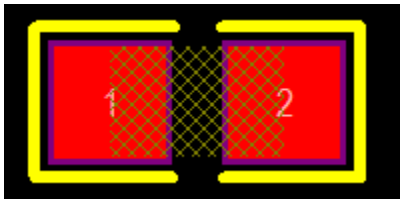




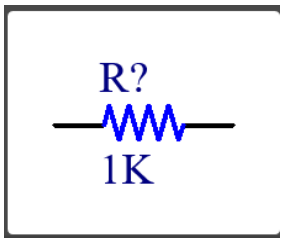
2. Capacitors (0.1uF,1uF,22pF)

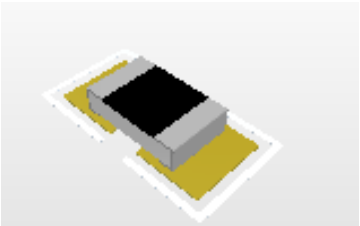
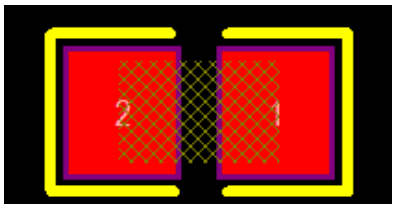




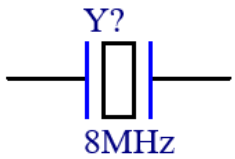


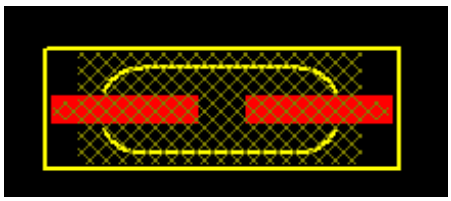
3. Resistors (10k ohms,1k ohms)



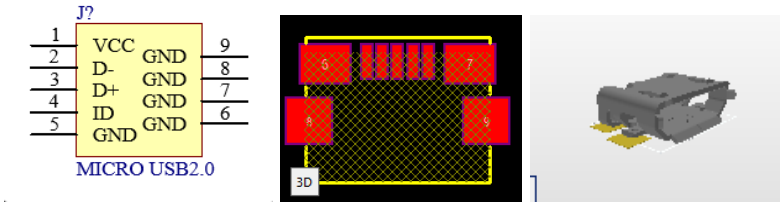



4. Crystal Oscillators (32.768KHz,8Mhz)

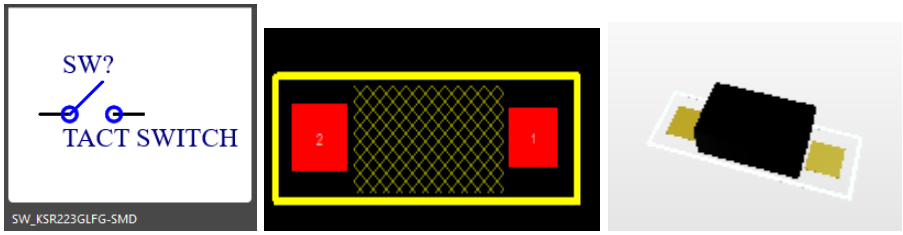




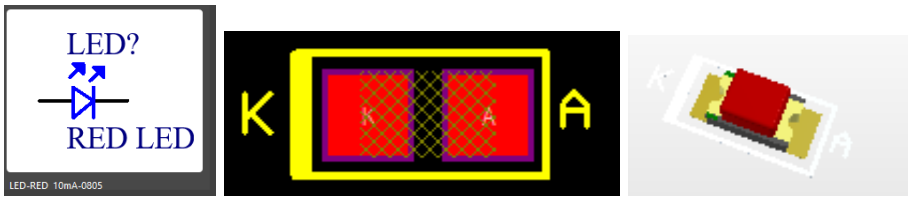
5. Connectors (2x3, 20x1, micro USB)



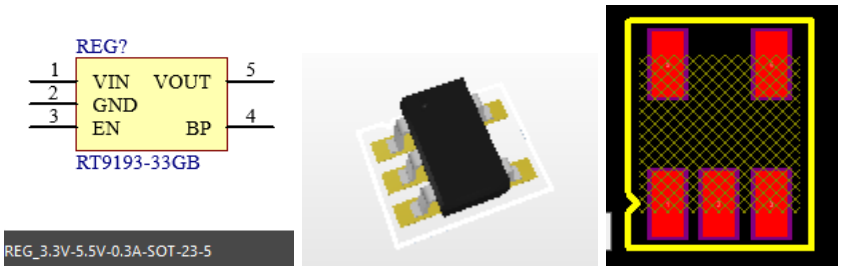
6. Switch



7. LEDs (Red)



8. Regulator (5v-3.3v)

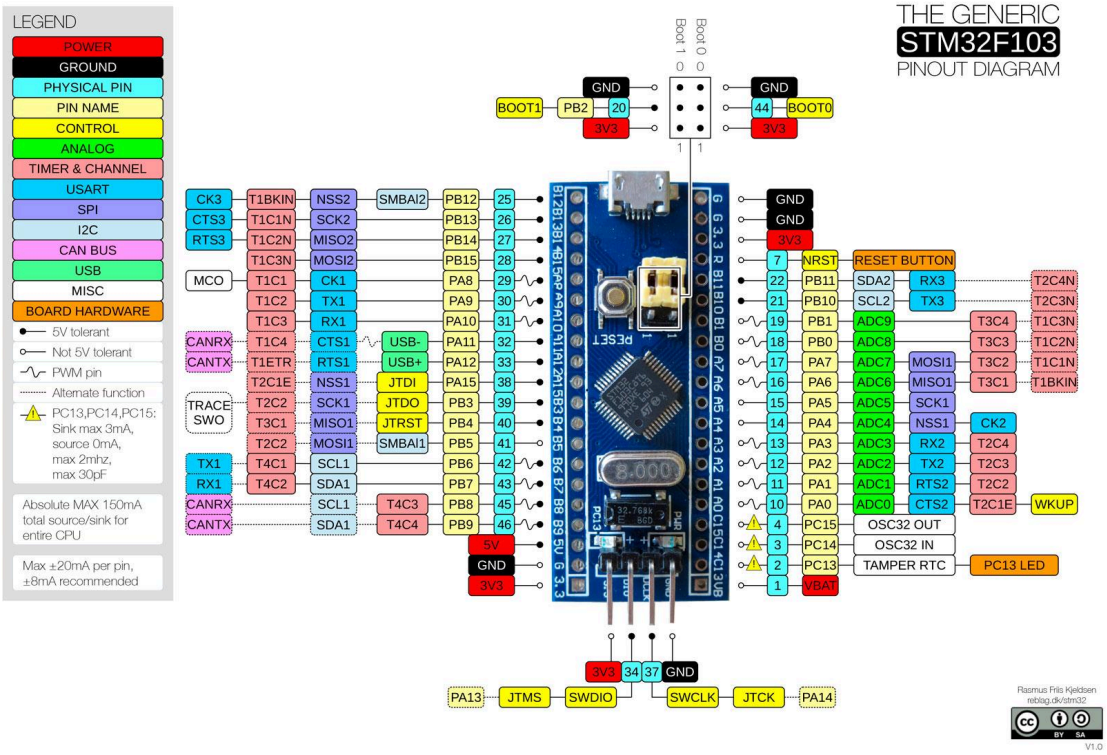


4.0 METHODOLOGY/ PROCEDURE

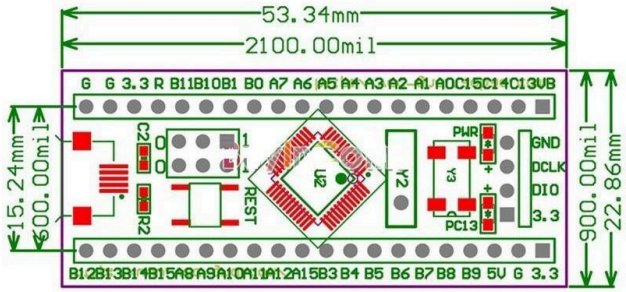
1. Create the components (symbol and footprint) in a library or download a library with the components needed and compile the library.
2. Create a new project, add new schematic and PCB document
3. On the schematic page, place the components and connect according to the circuit diagram for stm32f103
4. Annotate the components on the schematic page
5. Import the schematic to the PCB document
6. Resize the PCB to the appropriate size
7. Place the components properly on the PCB board
8. Make the connection using wires(track) and vias on the PCB board
9. For the ground connection,do a polygon pour
10. Prepare the job output file to be sent company for the PCB printing (and assembly)

5.0 Diagrams

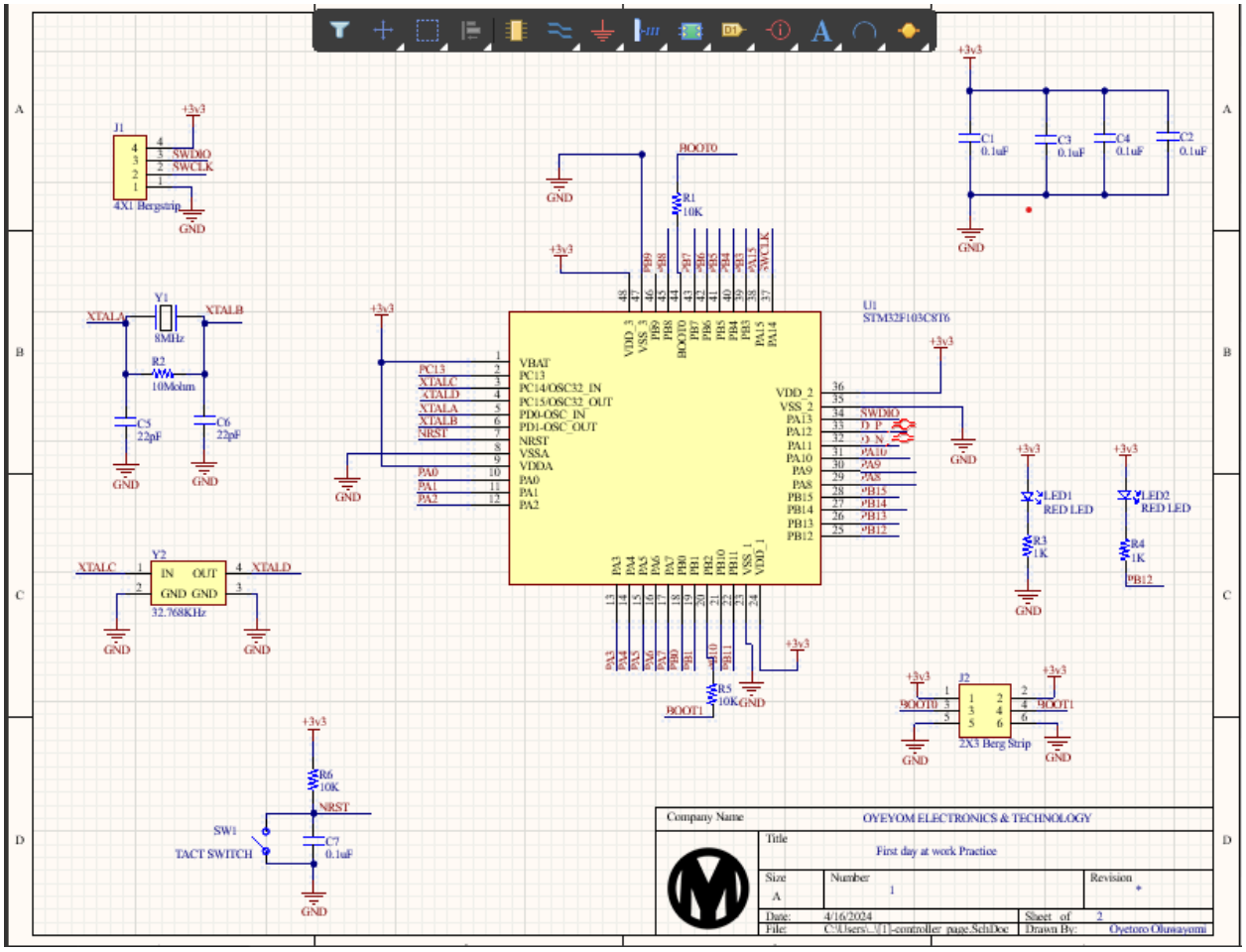
5.1 Circuit Diagram/ Pin out

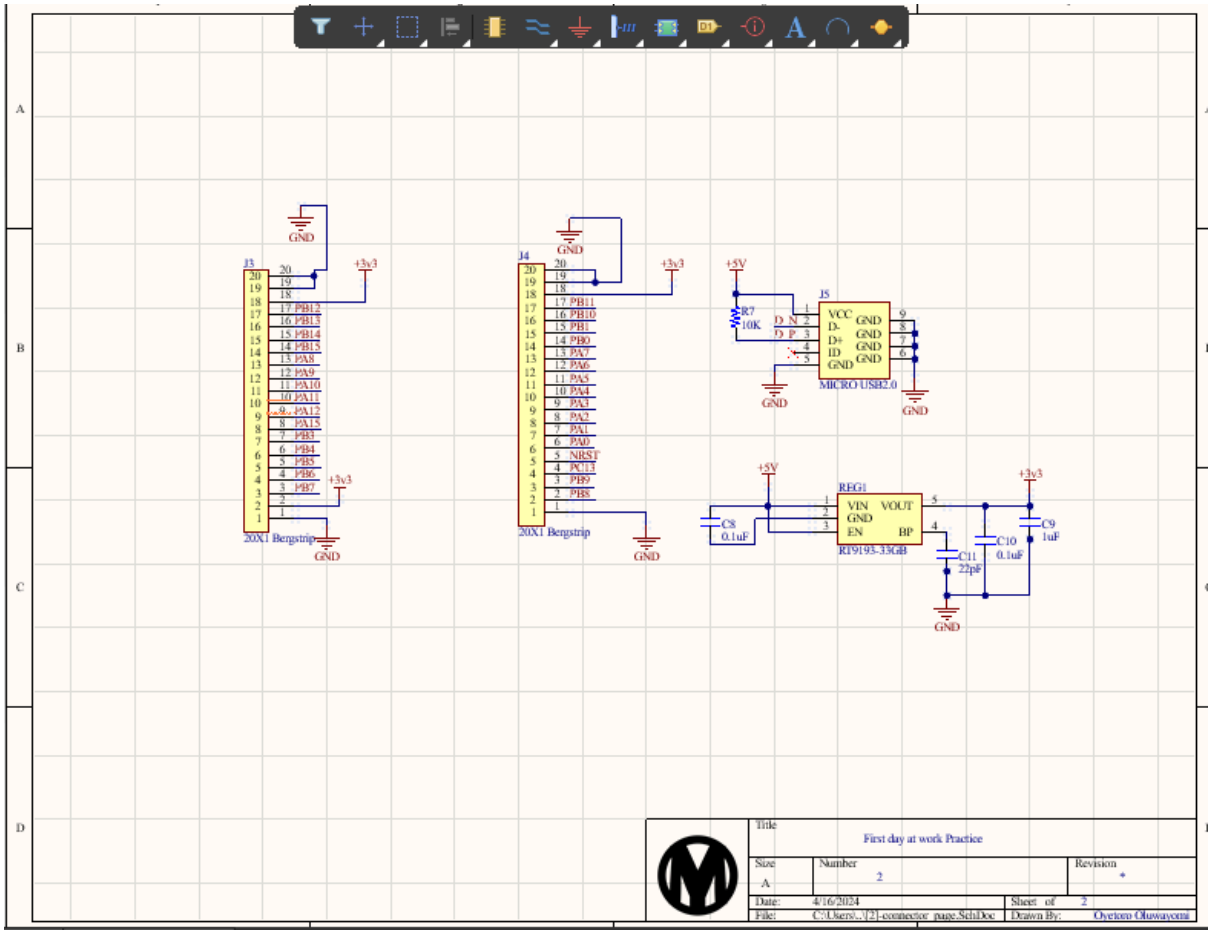


5.2 PCB board size

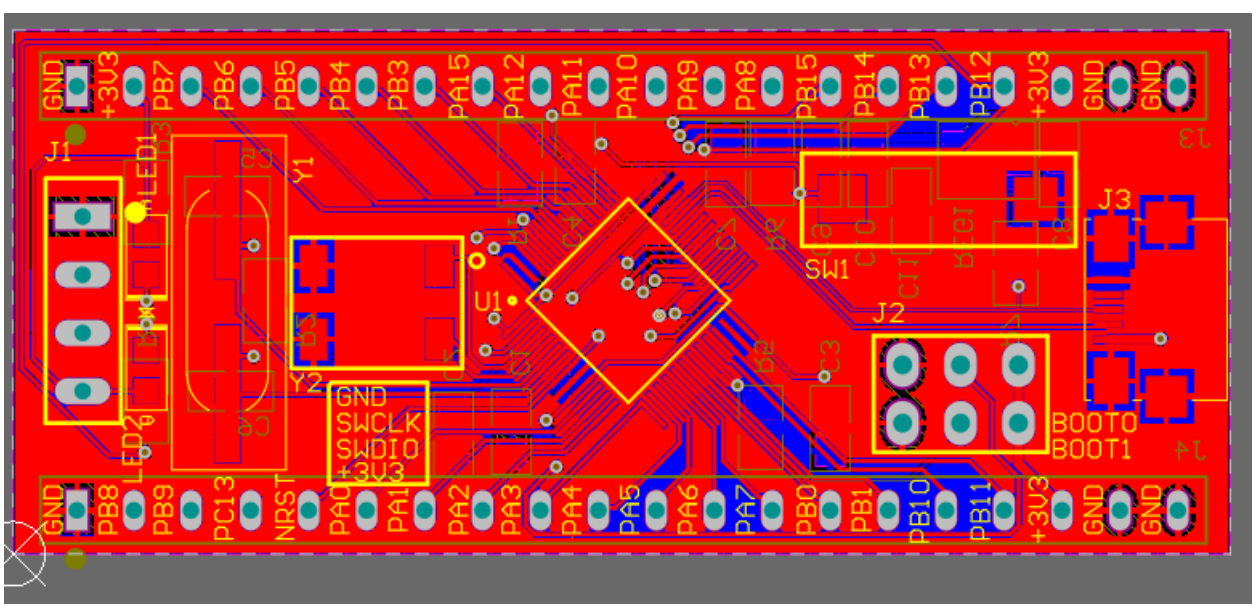


5.3 Schematic Diagram

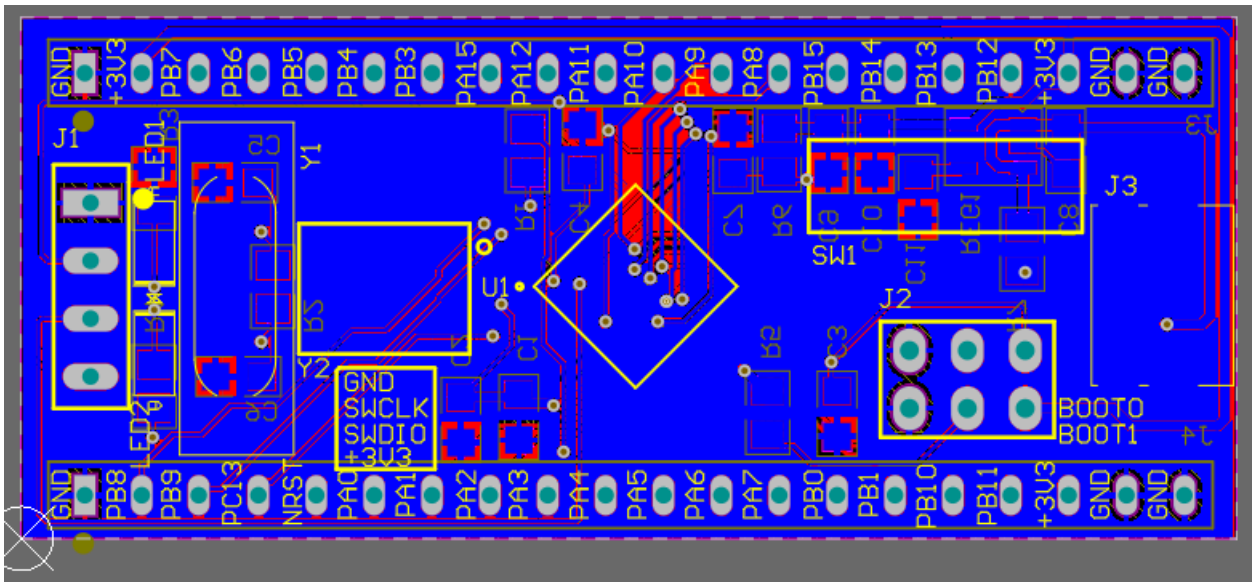




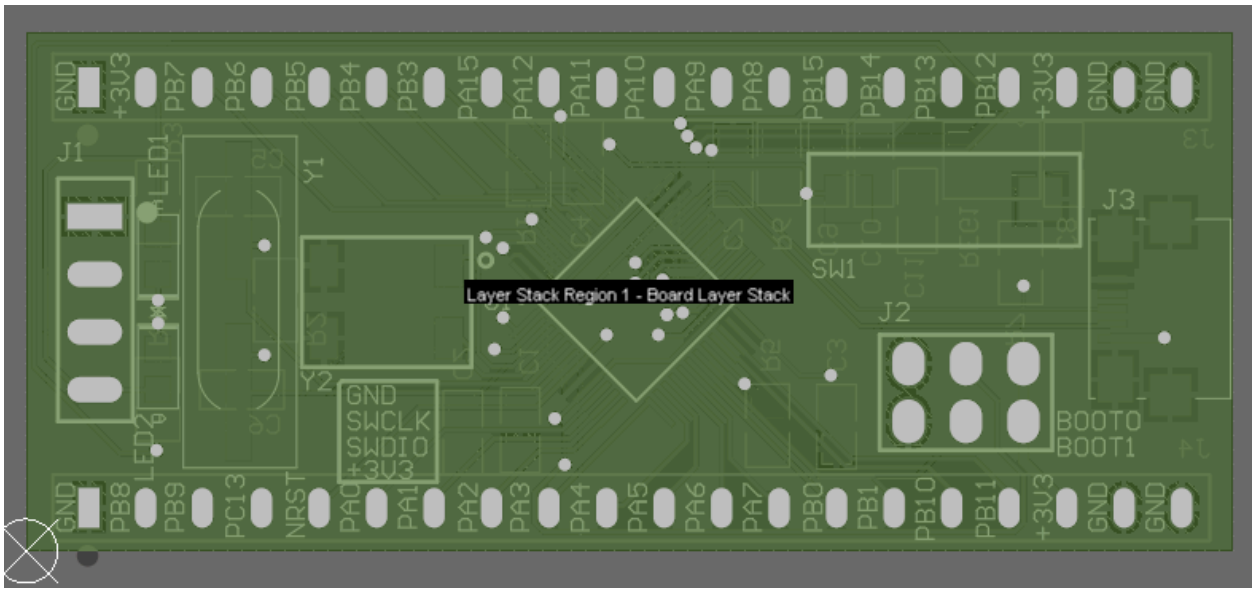
5.4 PCB Board



Top layer

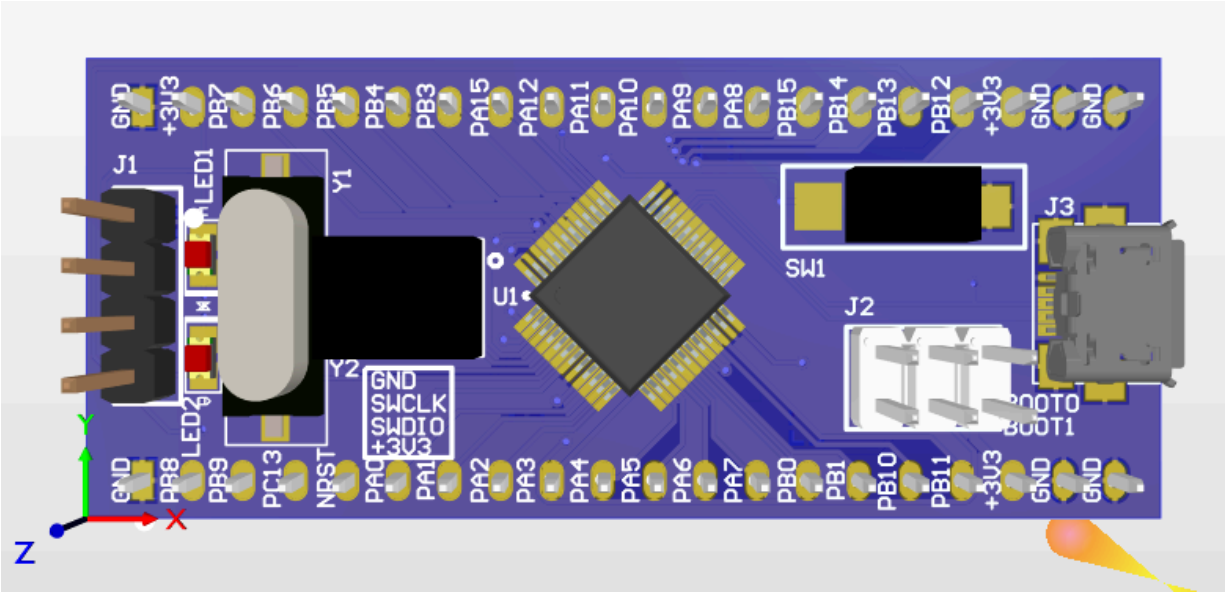


Bottom layer

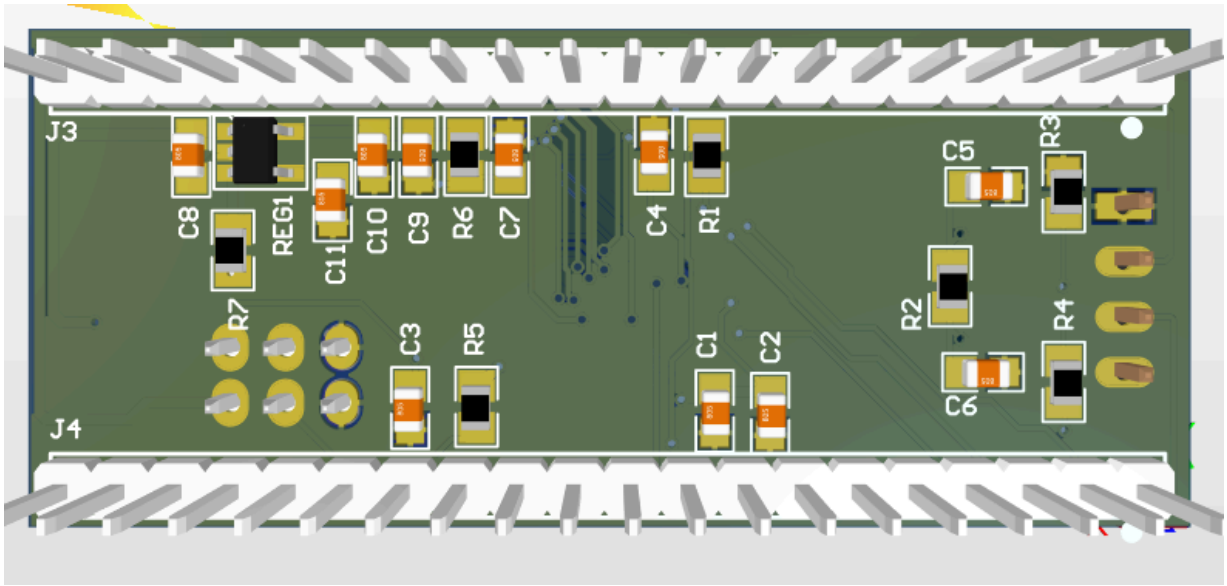


Layer stack

6.0 3D PCB BOARD



3D Top layer



3D Bottom Layer

7.0 CONCLUSION

The Printed Circuit board for STM32F103 was successfully designed and ready for printing.